PA-LSTM: Physics-Informed Acceleration-Aware LSTM
for Surgical Trajectory Latency Compensation in Medical Digital Twins

Peiang Miao

COMP5532 — Digital Twins & Virtual Human
Hong Kong Polytechnic University

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Outline

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Problem: Medical Digital Twin vs. Network Latency

- **Digital twin** in robotic surgery: virtual replica of instrument motion for navigation, training, teleoperation.
- **Challenge:** end-effector streams over WAN/5G incur **100–500 ms** delay $\Rightarrow$ twin shows "past" pose, not current.
- **Need: latency compensation** — predict future positions from a short history to resynchronise the twin.
- **Why 500 ms is hard:** at 30 Hz this is a **15-step** multi-step forecast; vanilla LSTM-style rollouts tend to **accumulate error**, drift in **phase**, and over-smooth rapid manoeuvres.

Preliminary Study: Why Not Mamba?

- **Deployment constraint:** surgical digital twins updates must cost $< 10\,\mathbf{ms}$ per inference on CPU to keep synchronisation.
- **Observation:** SSMS/Mamba models exceed this budget. Their selective scan is inherently causal/sequential on CPU, limiting parallelisation.
- **MSMamba:** Adding a speed-based gate didn't help (the network essentially turned it off).

Model	**Mean (ms)**	**Mean < 10 ms?**
Mamba	14.95	No
MSMamba	16.23	No
Vanilla LSTM	1.09	Yes
PA-LSTM (Ours)	**1.30**	Yes

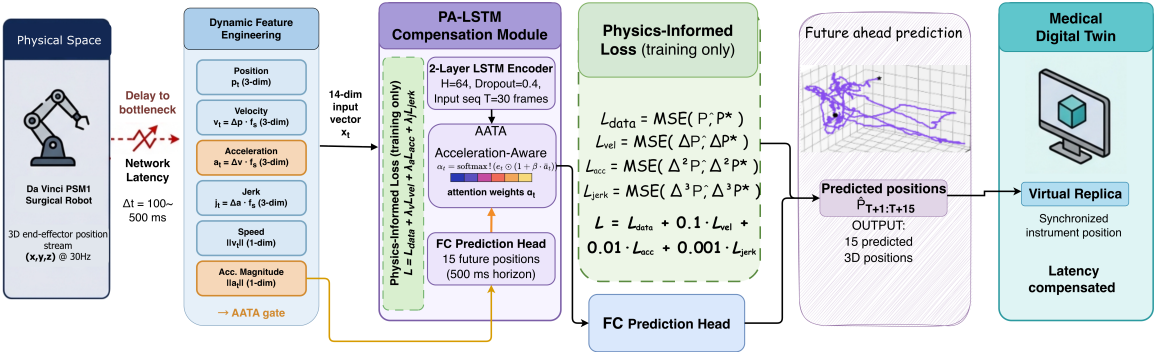
Takeaway: We need a **lightweight**, **physics-grounded** selective mechanism for LSTMs, rather than a heavier SSM scan on CPU.

PA-LSTM: Three Design Pillars

1. **AATA** (Acceleration-Aware Temporal Attention): modulate attention with normalised **acceleration magnitude** — like selective emphasis, but cheap.
2. **14-D features:** position, velocity, acceleration, jerk, speed, $|\mathbf{a}|$ — inform critical frames.
3. **Physics-informed loss:** match finite-difference velocity / acceleration / jerk to ground truth (minimum-jerk spirit).

System Pipeline (Overview)

PA-LSTM: Physics-Informed Acceleration-Aware LSTM for Surgical Trajectory Latency Compensation



AATA: Acceleration-Aware Temporal Attention

- **What AATA fixes:** standard temporal attention learns relevance from hidden states, but has no **kinematic grounding**.
- **Key signal:** acceleration magnitude $m_t = \|\mathbf{a}_t\|$ highlights direction reversals / braking moments that constrain the near future.
- **Cheap and derivable:** m_t is computed from finite differences of positions (no extra sensors).
- **Normalisation:** within a look-back window, $\tilde{a}_t \in [0, 1]$ rescales m_t into a stable physics gate.
- **Physics-modulated attention:** attention weights are amplified for high-acceleration frames, yielding a context vector $\mathbf{c} = \sum_t \alpha_t \mathbf{h}_t$.
- **Prediction head:** FC maps $\mathbf{c}$ to the next **15** 3D positions (**500 ms** horizon).

AATA: Core Equations & Overhead

Energy scores: $e_t = \mathbf{W}_a \tanh(\mathbf{W}_h \mathbf{h}_t + b)$

Normalised acc. magnitude: $\tilde{a}_t \in [0, 1]$

$$\alpha_t = \mathrm{softmax}\big(\{e_\tau\,(1 + \beta^+ \tilde{a}_\tau)\}\big)_t, \quad \beta^+ = \mathrm{softplus}(\beta), \quad \mathbf{c} = \sum_t \alpha_t \mathbf{h}_t$$

- If $\beta^+ \approx 0$, AATA **reduces to standard attention**.
- **Learnable modulation strength:** $\beta^+ = \mathrm{softplus}(\beta)$ keeps the acceleration gate non-negative and stable.
- **Inference overhead:** only element-wise modulation. Adds about **0.21 ms** over Vanilla LSTM on CPU (**1.09 →1.30 ms**).

Physics-Informed Training: PINN-style Minimum-Jerk Loss

$$\mathcal{L} = \mathcal{L}_{\text{data}} + 0.1\,\mathcal{L}_{\text{vel}} + 0.01\,\mathcal{L}_{\text{acc}} + 0.001\,\mathcal{L}_{\text{jerk}}$$

- **Innovation:** physics terms are **soft regularisers** (PINN-style), enforcing kinematic consistency via finite-difference velocity/acceleration/jerk constraints.
- **Minimum-jerk intuition:** smooth voluntary motion is preferred; physically implausible changes are penalised.
- **Training-only cost:** no additional inference computation; physical priors improve **15-step** prediction stability.

Impact: Ablations show these constraints significantly reduce error at the difficult 500 ms horizon.

Experimental Setup

- **Dataset:** JIGSAWS Suturing dataset (da Vinci PSM1 trajectories).
- **Data format:** 30 Hz time series of 3D positions (x, y, z).
- **Task:** 1 s history (30 frames) $\rightarrow$ predict up to 500 ms future (15 frames).
- **Evaluation:**
  - Subject-based cross-validation (generalisation to unseen surgeons).
  - Metric: **RMSE** (Root Mean Squared Error) across prediction steps.

What is RMSE? (Evaluation Standard)

RMSE (Root Mean Squared Error) measures average prediction error magnitude:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \|\hat{\mathbf{y}}_i - \mathbf{y}_i\|_2^2}$$

- **Here:** $\mathbf{y}_i \in \mathbb{R}^3$ is the ground-truth 3D position, $\hat{\mathbf{y}}_i$ is the predicted 3D position.
- **Aggregation:** errors are averaged over all predicted frames and all test windows (multi-step horizon).
- **Unit & meaning:** same unit as position (mm in our plots). **Lower is better**.
- **Why RMSE:** standard regression metric that penalises large deviations more strongly (important for safety-critical tracking).

Results: Accuracy and Real-Time Feasibility

RMSE (mm) Across Latency Conditions

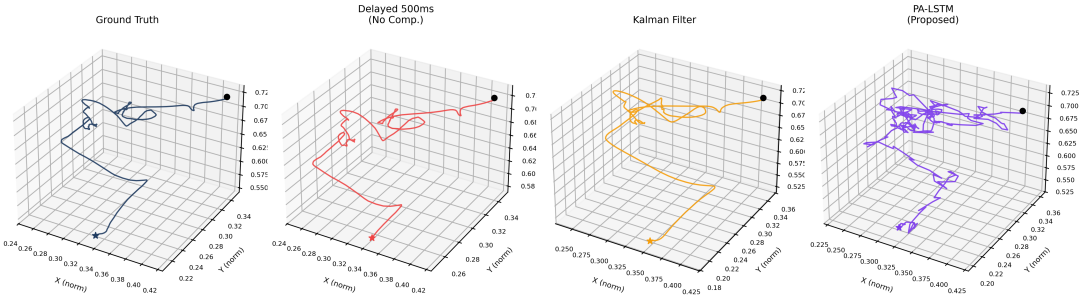
Method	100 ms	200 ms	500 ms	Params	CPU Lat.
Delayed (No Comp.)	0.1355	0.2634	0.5917	—	—
Kalman Filter	0.2179	0.3419	0.6994	—	—
Vanilla LSTM	**0.0030**	**0.0053**	0.0148	211,373	1.09 ms
Attention-LSTM	0.0035	0.0060	0.0153	211,502	1.15 ms
PA-LSTM w/o PI-Loss	0.0046	0.0060	0.0139	87,087	1.30 ms
PA-LSTM (Ours)	0.0045	0.0059	**0.0137**	**87,087**	**1.30 ms**

- **Short horizons (100/200 ms):** Simple LSTM baselines remain strong.
- **Long horizon (500 ms): PA-LSTM** excels by improving **rollout stability** via AATA + physics regularisation.
- **Efficiency:** Achieves SOTA using **fewer parameters** (87K) and maintains **real-time CPU speed** (1.30 ms).

Qualitative Results: 3D Trajectories (500 ms Horizon)

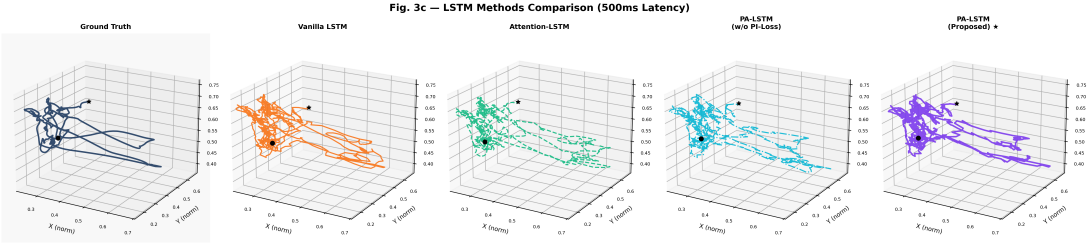
- PA-LSTM maintains phase and closely aligns with the Ground Truth at direction reversals, while the Kalman filter overshoots.

Fig. 3 — 3D Surgical Trajectory Comparison (500ms Latency)



Neural Methods Compared (500 ms)

- Vanilla LSTM and Attention-LSTM are competitive at shorter horizons but degrade with long-horizon accumulation.
- **Adding physics loss** reduces implausible motion changes across the prediction steps.
- **PA-LSTM (proposed)** produces the most faithful trajectory shape at 500 ms.



Conclusion

- **Feasibility insight:** Mamba-class models are currently **too slow on CPU** for embedded high-frequency surgical twins.
- **Method contribution:** PA-LSTM uses AATA to provide **cheap, physics-grounded selectivity** with minimal inference overhead.
- **Stability contribution:** Physics-informed (minimum-jerk style) training loss radically improves **multi-step** rollout stability.
- **Outcome:** Best 500 ms accuracy with a **1.30 ms** CPU inference latency, perfectly fitting the digital twin synchronisation budget.

Thank you.

Questions?